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Inventor(s)

LIM; Sun Ho et al.

ORTHOGONAL-AXIS POWER TRANSMISSION DEVICE

Abstract

Provided is an orthogonal-axis power transmission device. The orthogonal-axis power transmission device according to an embodiment of the present inventive concept includes: a main structure assembly forming an exterior structure and having a space portion formed therein, the space portion being sealed; an input shaft assembly connected to one side of the main structure assembly, and to which rotational power of a motor is input; an output shaft assembly connected to the other side of the main structure assembly intersecting the input shaft assembly, and from which the rotational power of the motor is output; and a link assembly for changing a power transmission direction arranged to operate in the space portion of the main structure assembly to prevent propagation of noise or vibration, connected to the input shaft assembly and the output shaft assembly within the main structure assembly by a link method, and changing the power transmission direction from the input shaft assembly to the output shaft assembly.

Inventors: LIM; Sun Ho (Asan-si, Chungcheongnam-do, KR), RYU; Min Ho (Cheonan-si, Chungcheongnam-do, KR)

Applicant: MATINPROUT CO., LTD. (Seoul, KR)

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Background/Summary

TECHNICAL FIELD

[0001] The present inventive concept relates to an orthogonal-axis power transmission device, and more particularly, to an orthogonal-axis power transmission device which may change a power transmission direction to an orthogonal axis while reducing noise and vibration, also have a compact structure that significantly reduces manufacturing costs compared to the related art, and furthermore enable manufacturing in a size suitable for an intended use so as to be widely applied to various industrial machines.

BACKGROUND ART

[0002] Power transmission devices may be largely classified into a rack and pinion that changes a rotational motion to a linear motion or a linear motion to a rotational motion, and a gear train that changes a rotation speed and torque while transferring only a rotational motion. The power transmission device are widely used for various industrial machines including semiconductor equipment, flat display equipment such as LCD, PDP, or OLED, and the like.

[0003] Meanwhile, for power transmission devices that are already applied or prepared to be applied to various industrial machines, the power transmission direction is generally linear, that is, along a straight axis. A linear shaft power transmission device adopts a method of using a reducer directly connected to a motor, so although the installation height is somewhat large, the device has the advantage of a very simple structure. Therefore, the linear shaft power transmission device is widely used.

[0004] However, there are times when the power transmission direction needs to be changed to an orthogonal axis instead of the existing straight axis due to a small installation space or interference with surrounding structures or devices. As such, when the power transmission direction needs to be used in an orthogonal axis, it is common to use a bevel gear.

[0005] Nevertheless, the bevel gear is a type of gear that operates as two gears are directly meshed with each other at a 45-degree angle with friction, so noise and vibration caused by the frictional motion are bound to be severe. In order to reduce this noise and vibration, there is a problem that manufacturing costs inevitably increase because expensive, precisely manufactured bevel gears are used.

[0006] Furthermore, considering the fact that it is practically impossible to manufacture large bevel gears over a certain size, there is a problem that there are some limitations in changing the power transmission direction to an orthogonal axis using a bevel gear.

DESCRIPTION OF EMBODIMENTS

Technical Problem

[0007] It is an objective of the present inventive concept to provide an orthogonal-axis power transmission device which may change a power transmission direction to an orthogonal axis while reducing noise and vibration, also have a compact structure that significantly reduces manufacturing costs compared to the related art, and furthermore enable manufacturing in a size

suitable for an intended use so as to be widely applied to various industrial machines.

Advantageous Effects of Disclosure

[0008] According to the present inventive concept, a power transmission direction may be changed to an orthogonal axis with reduced noise and vibration and also due to a compact structure, manufacturing costs may be significantly reduced compared to the related art, and furthermore, manufacturing in a size suitable for an intended use may be possible, thereby enabling a wide application to various industrial machines.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a perspective view of an orthogonal-axis power transmission device according to an embodiment of the present inventive concept.

[0010] FIG. 2 is a side view of FIG. 1.

[0011] FIG. 3 is a partially cut-away perspective view of FIG. 1.

[0012] FIG. 4 is a cross-sectional view of FIG. 2.

[0013] FIG. 5 is a partially cut-away perspective view of a main structure assembly.

[0014] FIG. 6 is a partially exploded perspective view of the main structure assembly.

[0015] FIG. 7 is a partially cut-away perspective view of an input shaft assembly.

[0016] FIG. 8 is a partially exploded perspective view of the input shaft assembly.

[0017] FIG. 9 is a partially cut-away perspective view of an output shaft assembly.

[0018] FIG. 10 is a partially exploded perspective view of the output shaft assembly.

[0019] FIG. 11 is a perspective view of a link assembly for changing a power transmission direction.

[0020] FIG. 12 is an exploded perspective view of the link assembly for changing a power transmission direction.

[0021] FIG. 13 is an enlarged perspective view of a main link member.

[0022] FIGS. 14A to 14F are views showing states of the link assembly for changing a power transmission direction rotated for each angle.

[0023] FIG. 15 is a perspective view of a link assembly for changing a power transmission direction applied to an orthogonal-axis power transmission device according to another embodiment of the present inventive concept.

BEST MODE

[0024] According to an aspect of the present inventive concept, an orthogonal-axis power transmission device includes: a main structure assembly forming an exterior structure and having a space portion formed therein, the space portion being sealed; an input shaft assembly connected to one side of the main structure assembly, and to which rotational power of a motor is input; an output shaft assembly connected to the other side of the main structure assembly intersecting the input shaft assembly, and from which the rotational power of the motor is output; and a link assembly for changing a power transmission direction arranged to operate in the space portion of the main structure assembly to prevent propagation of noise or vibration, connected to the input shaft assembly and the output shaft assembly within the main structure assembly by a link method, and changing the power transmission direction from the input shaft assembly to the output shaft assembly.

[0025] The link assembly for changing a power transmission direction may change the power transmission direction input from the input shaft assembly, by 90 degrees, and transmit an output to the output shaft assembly in one-to-one correspondence, the link assembly for changing a power transmission direction may include a main link member, an input shaft link member having one side connected to the input shaft assembly and the other side connected to one side of the main link

member to be rotatable relative to each other, and an output shaft link member having one side connected to the output shaft assembly and the other side connected to the other side of the main link member to be rotatable relative to each other.

[0026] The link assembly for changing a power transmission direction further may include an input shaft connection portion connecting the main link member and the input shaft link member to be rotatable relative to each other, and an output shaft connection portion connecting the main link member and the output shaft link member to be rotatable relative to each other.

[0027] The input shaft connection portion may include an input shaft sub-link having one side connected to the input shaft link member and the other side connected to the main link member, a first input shaft bearing arranged in a connection area between the input shaft sub-link and the input shaft link member, an input shaft power transmission pin integrally connected to the input shaft link member, the input shaft sub-link, and the first input shaft bearing and transmitting power from a side of the input shaft link member to the input shaft sub-link, and a second input shaft bearing arranged in a connection area between the input shaft sub-link and the main link member.

[0028] The input shaft connection portion further may include a plurality of input shaft stop rings provided to fix a position of the input shaft sub-link or the input shaft power transmission pin.

[0029] The output shaft connection portion may include an output shaft sub-link having one side connected to the output shaft link member and the other side connected to the main link member, a first output shaft bearing arranged in a connection area between the output shaft sub-link and the output shaft link member, an output shaft power transmission pin integrally connected to the output shaft link member, the output shaft sub-link, and the first output shaft bearing and transmitting power from a side of the output shaft link member to the output shaft sub-link, and a second output shaft bearing arranged in a connection area between the output shaft sub-link and the main link member.

[0030] The output shaft connection portion further may include a plurality of output shaft stop rings provided to fix a position of the output shaft sub-link or the output shaft power transmission pin.

[0031] The input shaft connection portion and the output shaft connection portion have the same structure.

[0032] The main link member may include a disc type link body manufactured in a disc shape and in which first and second link connection portions, to which the input shaft link member and the output shaft link member are respectively connected, may be formed on both sides in opposite directions, at least one weight reducer formed in a center area of the disc type link body by passing therethrough, and reducing weight of the disc type link body, and at least one inertia moment balancer provided in side regions of the disc type link body adjacent to the input shaft link member and the output shaft link member to balance moment of inertia during a rotational motion of the disc type link body.

[0033] The main link member is an S-type link member in an S shape, and the input shaft link member and the output shaft link member are provided in the same shape as a Y-type link member in a Y shape.

[0034] The input shaft link member is key-coupled to the input shaft assembly by an input shaft key, and the output shaft link member is key-coupled to the output shaft assembly by an output shaft key.

[0035] The main structure assembly may include a structure body in which the space portion may be formed, and on one side of which an input shaft assembly connection portion, to which the input shaft assembly is connected, may be formed, a structure cover on which an output shaft assembly connection portion, to which the output shaft assembly is connected, is provided, the structure cover being connected to the structure body, and a plurality of structure cover fastening members detachably fastening the structure cover and the structure body.

[0036] The main structure assembly may include a structure cover O-ring arranged between the

structure cover and the structure body and hermetically sealing between the structure cover and the structure body, a structure cover dowel pin arranged between the structure cover and the structure body and aligning a position of the structure cover with respect to the structure body, a front cover shielding one side front opening of the structure body, a plurality of front cover fastening members detachably fastening the front cover to the structure body, and a front cover O-ring arranged between the front cover and the structure body and hermetically sealing between the front cover and the structure body.

[0037] The input shaft assembly may include an input shaft structure connected to the one sides of the main structure assembly, a plurality of input shaft fixing bolts to fix the input shaft structure to the main structure assembly, an input shaft adapter rotated by the motor, having one side connected to the link assembly for changing a power transmission direction, and inputting the rotational power of the motor to the link assembly for changing a power transmission direction, an input shaft adapter bearing arranged in the input shaft structure and supporting a rotational motion of the input shaft adapter, and an input shaft adapter bearing preload plate forming a preload on the input shaft adapter bearing.

[0038] The input shaft assembly may include a motor installation plate on which the motor is mounted, a clamp fastening a motor shaft of the motor to the input shaft adapter, and a clamp tightening bolt tightening the clamp.

[0039] The output shaft assembly may include an output shaft structure connected to the other side of the main structure assembly, a plurality of output shaft fixing bolts fixing the output shaft structure to the main structure assembly, an output shaft adapter connected to the other side of the link assembly for changing a power transmission direction and outputting power transmitted from the link assembly for changing a power transmission direction, an output shaft adapter bearing arranged in the output shaft structure and supporting a rotational motion of the output shaft adapter, and an output shaft adapter bearing preload plate forming a preload on the output shaft adapter bearing.

Mode of Disclosure

[0040] The attached drawings for illustrating preferred embodiments of the present inventive concept are referred to in order to gain a sufficient understanding of the present inventive concept, the merits thereof, and the objectives accomplished by the implementation of the present inventive concept.

[0041] Hereinafter, the present inventive concept will be described in detail by explaining preferred embodiments of the inventive concept with reference to the attached drawings. Like reference numerals in the drawings denote like elements.

[0042] FIG. 1 is a perspective view of an orthogonal-axis power transmission device according to an embodiment of the present inventive concept. FIG. 2 is a side view of FIG. 1. FIG. 3 is a partially cut-away perspective view of FIG. 1. FIG. 4 is a cross-sectional view of FIG. 2. FIG. 5 is a partially cut-away perspective view of a main structure assembly. FIG. 6 is a partially exploded perspective view of the main structure assembly. FIG. 7 is a partially cut-away perspective view of an input shaft assembly. FIG. 8 is a partially exploded perspective view of the input shaft assembly. FIG. 9 is a partially cut-away perspective view of an output shaft assembly. FIG. 10 is a partially exploded perspective view of the output shaft assembly. FIG. 11 is a perspective view of a link assembly for changing a power transmission direction. FIG. 12 is an exploded perspective view of the link assembly for changing a power transmission direction. FIG. 13 is an enlarged perspective view of a main link member. FIGS. 14A to 14F are views showing states of the link assembly for changing a power transmission direction rotated for each angle.

[0043] Referring to the drawings, an orthogonal-axis power transmission device **100** according to an embodiment of the present inventive concept may change a power transmission direction to an orthogonal axis, as shown in FIG. 2, while reducing noise and vibration, also have a compact structure that significantly reduces manufacturing costs compared to the related art, and

furthermore enable manufacturing in a size suitable for an intended use so as to be widely applied to various industrial machines.

[0044] The orthogonal-axis power transmission device **100** according to an embodiment of the present inventive concept, which may provide the above effects, may include a main structure assembly **110** forming an exterior structure, and have a structure in which an input shaft assembly **130**, an output shaft assembly **150**, and a link assembly **170** for changing a power transmission direction are coupled to the main structure assembly **110** for each position.

[0045] The main structure assembly **110** has a box structure forming an exterior structure as illustrated in FIGS. **1** to **6**. The main structure assembly **110** has a space portion **112** therein which is sealed so that the link assembly **170** for changing a power transmission direction may be disposed.

[0046] As the link assembly **170** for changing a power transmission direction is disposed inside the space portion **112** that is sealed within the main structure assembly **110**, and the link assembly **170** for changing a power transmission direction operates as illustrated in FIGS. **14A** to **14F**, in changing the power transmission direction to the orthogonal axis as illustrated in FIG. **2**, noise and vibration may be significantly reduced compared to the related art.

[0047] The main structure assembly **110** may include a structure body **111** and a structure cover **114** which are disassembled and assembled from each other.

[0048] The structure body **111** has a box-shaped structure in which the space portion **112** is formed, and on one side of which an input shaft assembly connection portion **113**, to which the input shaft assembly **130** is connected, is formed. The input shaft assembly connection portion **113** has a hole shape, and the input shaft assembly **130** is hermetically connected to the input shaft assembly connection portion **113**.

[0049] An opening **125**, which has a similar shape as the input shaft assembly connection portion **113** having a hole shape, is formed on one front side of the structure body **111**, and a front cover **121** is provided to shield the opening **125** on the front side of the structure body **111**.

[0050] The front cover **121** is fastened to a corresponding position by a front cover fastening member **122**. In other words, the front cover fastening member **122** detachably fastens the front cover **121** to the structure body **111**. As the front cover **121** is fastened by the front cover fastening member **122**, the opening **125** in the front side of the structure body **111** may be shielded.

[0051] In this state, a front cover O-ring **123** is used between the front cover **121** and the structure body **111** to maintain the seal between the front cover **121** and the structure body **111**. Accordingly, leakage of internal lubricant through the opening **125** in the front side of the structure body **111** may be prevented.

[0052] The structure cover **114** is a structure that is detachably connected to the structure body **111**. A plurality of structure cover fastening members **116** are used for attachment and detachment of the structure cover **114**. The structure cover fastening members **116** is a means for detachably fastening the structure cover **114** and the structure body **111**. As such, as the structure cover **114** is detachable, the link assembly **170** for changing power transmission direction may be installed in the space portion **112** within the structure body **111**, and furthermore, lubrication and maintenance and repair may be possible.

[0053] An output shaft assembly connection portion **115**, to which the output shaft assembly **150** is connected, is provided on the structure cover **114**. The output shaft assembly connection portion **115** has a protruding structure, unlike the input shaft assembly connection portion **113**.

[0054] A structure cover O-ring **117** is arranged between the structure cover **114** and the structure body **111**. The structure cover O-ring **117** is arranged between the structure cover **114** and the structure body **111** and hermetically seals between the structure cover **114** and the structure body **111**. The structure cover O-ring **117** may prevent leakage of internal lubricant between the structure cover **114** and the structure body **111**.

[0055] A structure cover dowel pin **118** is further arranged between the structure cover **114** and the

structure body **111**. The structure cover dowel pin **118** aligns the position of the structure cover **114** with respect to the structure body **111**. Accordingly, the disassembly and assembly, especially assembly, of the structure cover **114** becomes easy.

[0056] As illustrated in FIGS. **1** to **4** and FIGS. **7** and **8**, the input shaft assembly **130** is a part which is connected to one side of the main structure assembly **110** and to which rotational power of a motor (not shown) is input.

[0057] The input shaft assembly **130** may include an input shaft structure **131** connected to one side of the main structure assembly **110**, and structures and components described below are connected by position through the input shaft structure **131**.

[0058] A plurality of input shaft fixing bolts **132** are used to fix the input shaft structure **131**. In other words, the input shaft fixing bolts **132** are means for fixing the input shaft structure **131** to the main structure assembly **110**.

[0059] An input shaft adapter **133** is arranged within the input shaft structure **131**. The input shaft adapter **133** is rotated by a motor and has one side connected to an input shaft link member **172** of the link assembly **170** for changing a power transmission direction, and inputs the power of the motor to the input shaft link member **172**. The input shaft link member **172** is key-coupled to the input shaft adapter **133** of the input shaft assembly **130** by an input shaft key **173**. Accordingly, the input shaft assembly **130** is prevented from spinning.

[0060] A clamp **137** for fastening a motor shaft of the motor to the input shaft adapter **133** is provided. The clamp **137** may be tightened by a clamp tightening bolt **138**.

[0061] An input shaft adapter bearing **134** is arranged around the input shaft adapter **133**. The input shaft adapter bearing **134** is arranged in the input shaft structure **131** and supports the rotational motion of the input shaft adapter **133**. In other words, the input shaft adapter bearing **134** facilitates the rotational motion of the input shaft adapter **133**.

[0062] An input shaft adapter bearing preload plate **135** is used to install the input shaft adapter bearing **134**. The input shaft adapter bearing preload plate **135** is a means for forming a preload on the input shaft adapter bearing **134**. In other words, the input shaft adapter bearing preload plate **135** increases bearing rotation precision by adjusting clearance of the input shaft adapter bearing **134** and also adjusts an assembly distance of the link assembly **170** for changing a power transmission direction.

[0063] A motor installation plate **136** is installed around the input shaft structure **131**. The motor installation plate **136** is a part where the motor that is not shown is installed. The motor installation plate **136** is fixed to a corresponding position by a motor installation plate fixing bolt **141**.

[0064] An input shaft oil seal **142** is provided in the input shaft assembly **130**. The input shaft oil seal **142** may prevent leakage of internal lubricant through the input shaft assembly **130**.

[0065] The output shaft assembly **150**, as illustrated in FIGS. **1** to **4** and FIGS. **9** and **10**, is a part which is connected to the other side of the main structure assembly **110** to intersect the input shaft assembly **130**, and through which the rotational power of the motor is output. Excluding the installation structure of the motor, the output shaft assembly **150** has a structure similar to the input shaft assembly **130**.

[0066] The output shaft assembly **150** may also include an output shaft structure **151** connected to the other side of the main structure assembly **110**, and structures and components described below are connected by position through the output shaft structure **151**.

[0067] A plurality of output shaft fixing bolts **152** are used for fixing the output shaft structure **151**. In other words, the output shaft fixing bolts **152** are means to fix the output shaft structure **151** to the main structure assembly **110**.

[0068] An output shaft adapter **153** is arranged within the output shaft structure **151**. The output shaft adapter **153** is a means which is connected to the other side of the link assembly **170** for changing a power transmission direction, that is, an output shaft link member **174** of the link assembly **170** for changing a power transmission direction and outputs power transmitted from the

link assembly **170** for changing a power transmission direction. Rotational power may be provided by connecting device via the output shaft adapter **153**.

[0069] The output shaft link member **174** may be key-coupled to the output shaft adapter **153** of the output shaft assembly **150** by an output shaft key **175**. Accordingly, the output shaft link member **174** is prevented from spinning.

[0070] An output shaft adapter bearing **154** for supporting the rotational motion of the output shaft adapter **153** is provided in the output shaft structure **151**. An output shaft adapter bearing preload plate **155** for forming a preload on the output shaft adapter bearing **154** is installed to be connected to the output shaft adapter bearing **154**. The output shaft adapter bearing preload plate **155** also provides the same function as the input shaft adapter bearing preload plate **135** described above. In other words, the output shaft adapter bearing preload plate **155** increases bearing rotation precision by adjusting clearance of the output shaft adapter bearing **154** and also adjusts an assembly distance of the link assembly **170** for changing a power transmission direction.

[0071] Meanwhile, the link assembly **170** for changing a power transmission direction, as illustrated in FIGS. **1** to **4** and FIGS. **11** and **12**, is arranged in the space portion **112** of the main structure assembly **110** to prevent propagation of noise or vibration, operating as illustrated in FIGS. **14A** to **14F**, for example, is connected to the input shaft assembly **130** and the output shaft assembly **150** inside the main structure assembly **110** by a link method, and changes the power transmission direction from the input shaft assembly **130** to the output shaft assembly **150**.

[0072] In particular, the link assembly **170** for changing a power transmission direction applied to the present embodiment changes the power transmission direction input from the input shaft assembly **130**, by 90 degrees, and transmits an output to the output shaft assembly **150** in one-to-one correspondence.

[0073] A link structure provides less noise and vibration like the bevel gear according to the related art. Furthermore, the link assembly **170** for changing a power transmission direction applied to the present embodiment, which is arranged in the sealed space portion **112** of the main structure assembly **110** and then operates, noise and vibration may be prevented from being propagated to the outside. Furthermore, the structure of the link assembly **170** for changing a power transmission direction, which adopts the link structure, is simple so that manufacturing costs may be reduced, and furthermore, the link assembly **170** for changing a power transmission direction may be manufactured in a desired size, unlike the bevel gear.

[0074] The link assembly **170** for changing a power transmission direction may include a main link member **171**, the input shaft link member **172** having one side connected to the input shaft assembly **130** and the other side connected to one side of the main link member **171** to be rotatable relative to each other, and the output shaft link member **174** having one side connected to the output shaft assembly **150** and the other side connected to the other side of the main link member **171** to be rotatable relative to each other.

[0075] The main link member **171**, as a link in the center portion of the link assembly **170** for changing a power transmission direction, forms a place where the input shaft link member **172** and the output shaft link member **174** are connected to opposite end regions thereof.

[0076] In the present embodiment, the main link member **171** may include a disc type link body **171a** in a disc shape, at least one weight reducer **171b** formed in the center area of the disc type link body **171a** by passing therethrough to reduce the weight of the disc type link body **171a**, and at least one inertia moment balancer **171c** provided in side regions of the disc type link body **171a** adjacent to the input shaft link member **172** and the output shaft link member **174** to balance the moment of inertia during the rotational motion of the disc type link body **171a**.

[0077] The disc type link body **171a** has a circular outer shape except portions that contact the input shaft link member **172** and the output shaft link member **174**. The disc type link body **171a** serves as a flywheel to implement a smooth rotation with the input shaft link member **172** and the output shaft link member **174**.

[0078] First and second link connection portions **171d** and **171e**, to which the input shaft link member **172** and the output shaft link member **174** are respectively connected, are formed on both sides of the disc type link body **171a**. The first and second link connection portions **171d** and **171e** are provided in a protruding shape, and are arranged in the opposite directions on both sides of the disc type link body **171a**.

[0079] The weight reducer **171b** is formed in the center area of the disc type link body **171a** to reduce the weight of the disc type link body **171a**. In a case according to the present embodiment, the weight reducer **171b** is formed in a circular through-hole shape.

[0080] The inertia moment balancer **171c** is provided in the side regions of the disc type link body **171a** and balances the moment of inertia during the rotational motion of the disc type link body **171a**. In the present embodiment, the inertia moment balancer **171c** is equally arranged in the side regions on both sides of the disc type link body **171a** adjacent to the input shaft link member **172** and the output shaft link member **174**. The inertia moment balancer **171c** may also in the form of a hole.

[0081] The input shaft link member **172** and the output shaft link member **174** coupled to the opposite end regions of the main link member **171** are provided in the same shape as a Y-type link member in a Y shape.

[0082] The input shaft link member **172**, as a Y-type link member, has one side connected to the input shaft assembly **130** and the other side connected to one side of the main link member **171** to be rotatable relative to each other. Furthermore, the output shaft link member **174**, as a Y-type link member, has one side connected to the output shaft assembly **150** and the other side connected to the other side of the main link member **171** to be rotatable relative to each other.

[0083] In addition, the link assembly **170** for changing a power transmission direction may further include an input shaft connection portion **180** and an output shaft connection portion **190**, as means for connecting the input shaft link member **172** and the output shaft link member **174** to the opposite end regions of the main link member **171** to be rotatable relative to each other. In the present embodiment, as the input shaft connection portion **180** and the output shaft connection portion **190** have the same structure, assembly and maintenance and repair become convenient. However, it is not necessary that the input shaft connection portion **180** and the output shaft connection portion **190** have the same structure, and thus, the scope of rights of the present inventive concept is not limited thereto.

[0084] First, the input shaft connection portion **180** is a means for connecting the main link member **171** and the input shaft link member **172** to be rotatable relative to each other, and includes various parts and units.

[0085] The input shaft connection portion **180** may include an input shaft sub-link **181** having one side connected to the input shaft link member **172** and the other side connected to the main link member **171**, a first input shaft bearing **182** arranged in a connection area between the input shaft sub-link **181** and the input shaft link member **172**, an input shaft power transmission pin **183** integrally connected to the input shaft link member **172**, the input shaft sub-link **181**, and the first input shaft bearing **182**, and transmitting power from the side of the input shaft link member **172** to the input shaft sub-link **181**, and a second input shaft bearing **184** arranged in a connection area between the input shaft sub-link **181** and the main link member **171**.

[0086] One end portion of the input shaft sub-link **181** is inserted into an open portion of the input shaft link member **172** as a Y-type link member, and the other end portion thereof is coupled to a hole or opening formed in an end portion of the main link member **171** as an S-type link member. As the input shaft sub-link **181** connects between the input shaft link member **172** and the main link member **171** to be rotatable relative to each other, the first input shaft bearing **182** and the second input shaft bearing **184** are installed in both end portions of the input shaft sub-link **181** to induce smooth rotations. In particular, in a case of the first input shaft bearing **182**, as one end portion of the input shaft sub-link **181** is inserted into the open portion of the input shaft link

member **172**, the input shaft power transmission pin **183** is further used in the first input shaft bearing **182**.

[0087] A plurality of input shaft stop rings **185** are further used in the input shaft connection portion **180**. The input shaft stop rings **185** are provided to fix the position of the input shaft sub-link **181** or the input shaft power transmission pin **183**. The input shaft stop rings **185** are of E-shaped rings.

[0088] Next, the output shaft connection portion **190** is a means for connecting the main link member **171** and the output shaft link member **174** to be rotatable relative to each other. The output shaft connection portion **190** includes various parts and units.

[0089] The output shaft connection portion **190** has substantially the same structure, function, and role as the input shaft connection portion **180**, except a location thereof. In other words, the output shaft connection portion **190** may include an output shaft sub-link **191** having one side connected to the output shaft link member **174** and the other side connected to the main link member **171**, a first output shaft bearing **192** arranged in a connection area between the output shaft sub-link **191** and the output shaft link member **174**, an output shaft power transmission pin **193** integrally connected to the output shaft link member **174**, the output shaft sub-link **191**, and the first output shaft bearing **192** and transmitting power from the side of the output shaft link member **174** to the output shaft sub-link **191**, and a second output shaft bearing **194** arranged in a connection area between the output shaft sub-link **191** and the main link member **171**.

[0090] A plurality of output shaft stop rings **195**, as E-shaped rings, are further used in the output shaft connection portion **190**. The output shaft stop rings **195** is provided to fix the position of the output shaft sub-link **191** or the output shaft power transmission pin **193**.

[0091] Hereinafter, the operation of the orthogonal-axis power transmission device **100** according to an embodiment of the present inventive concept is described.

[0092] As described above, when the motor that is not shown is mounted on the input shaft assembly **130** and driven, the rotational power of the motor is input through the input shaft assembly **130**, and then, while a power transmission direction is changed 90 degrees by the link assembly **170** for changing a power transmission direction, the rotational power may be transmitted through the output shaft assembly **150** in one-to-one correspondence. In other words, the power of the motor is output to the output shaft assembly **150** in one-to-one correspondence with only the direction thereof changed.

[0093] In this state, the state of the link assembly **170** for changing a power transmission direction being rotated by angle are as shown in FIGS. **14A** to **14F**. In other words, the states of the link assembly **170** for changing a power transmission direction being rotated by 0 degree, 60 degrees, 120 degrees, 180 degrees, 240 degrees, and 300 degrees are as shown in FIGS. **14A** to **14F**.

[0094] According to the present embodiment operating with the structure described above, an orthogonal-axis power transmission device may change a power transmission direction to the orthogonal axis while reducing noise and vibration, also have a compact structure that significantly reduces manufacturing costs compared to the related art, and furthermore enable manufacturing in a size suitable for an intended use so as to be widely applied to various industrial machines.

[0095] FIG. **15** is a perspective view of a link assembly for changing a power transmission direction applied to an orthogonal-axis power transmission device according to another embodiment of the present inventive concept.

[0096] Referring to the drawing, a link assembly **270** for changing a power transmission direction applied to the present embodiment may also include a main link member **271**, the input shaft link member **172** having one side connected to the input shaft assembly **130** and the other side connected to one side of the main link member **271** to be rotatable relative to each other, and the output shaft link member **174** having one side connected to the output shaft assembly **150** and the other side connected to the other side of the main link member **271** to be rotatable relative to each other.

[0097] The structures, functions, and roles of the input shaft link member 172 and the output shaft link member 174 are the same as those of the embodiment described above. Accordingly, redundant descriptions thereof are omitted.

[0098] Meanwhile, the main link member 271 applied to the present embodiment, as a link in the center portion of the link assembly 270 for changing a power transmission direction, forms a place where the input shaft link member 172 and the output shaft link member 174 are connected to opposite end regions thereof. In the present embodiment, the main link member 271 is provided as an S-type link member in an S shape. Even when the main link member 271 is an S-type link member, the effects of the present inventive concept may be provided.

[0099] While this inventive concept has been particularly shown and described with reference to preferred embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the inventive concept as defined by the appended claims. Therefore, the scope of the inventive concept is defined not by the detailed description of the inventive concept but by the appended claims, and all differences within the scope will be construed as being included in the present inventive concept.

Industrial Applicability

[0100] The present inventive concept relates to various machine tools requiring a rotation motion or a linear motion, industrial machinery devices, semiconductor or flat display manufacturing equipment, various logistics transfer equipment, and the like.

Claims

1. An orthogonal-axis power transmission device comprising: a main structure assembly forming an exterior structure and having a space portion formed therein, the space portion being sealed; an input shaft assembly connected to one side of the main structure assembly, and to which rotational power of a motor is input; an output shaft assembly connected to the other side of the main structure assembly intersecting the input shaft assembly, and from which the rotational power of the motor is output; and a link assembly for changing a power transmission direction arranged to operate in the space portion of the main structure assembly to prevent propagation of noise or vibration, connected to the input shaft assembly and the output shaft assembly within the main structure assembly by a link method, and changing the power transmission direction from the input shaft assembly to the output shaft assembly.
2. The orthogonal-axis power transmission device of claim 1, wherein the link assembly for changing a power transmission direction changes the power transmission direction input from the input shaft assembly, by 90 degrees, and transmits an output to the output shaft assembly in one-to-one correspondence, the link assembly for changing a power transmission direction comprises: a main link member; an input shaft link member having one side connected to the input shaft assembly and the other side connected to one side of the main link member to be rotatable relative to each other; and an output shaft link member having one side connected to the output shaft assembly and the other side connected to the other side of the main link member to be rotatable relative to each other.
3. The orthogonal-axis power transmission device of claim 2, wherein the link assembly for changing a power transmission direction further comprises: an input shaft connection portion connecting the main link member and the input shaft link member to be rotatable relative to each other; and an output shaft connection portion connecting the main link member and the output shaft link member to be rotatable relative to each other.
4. The orthogonal-axis power transmission device of claim 3, wherein the input shaft connection portion comprises: an input shaft sub-link having one side connected to the input shaft link member and the other side connected to the main link member; a first input shaft bearing arranged in a

connection area between the input shaft sub-link and the input shaft link member; an input shaft power transmission pin integrally connected to the input shaft link member, the input shaft sub-link, and the first input shaft bearing and transmitting power from a side of the input shaft link member to the input shaft sub-link; and a second input shaft bearing arranged in a connection area between the input shaft sub-link and the main link member.

5. The orthogonal-axis power transmission device of claim 4, wherein the input shaft connection portion further comprises a plurality of input shaft stop rings provided to fix a position of the input shaft sub-link or the input shaft power transmission pin.

6. The orthogonal-axis power transmission device of claim 3, wherein the output shaft connection portion comprises: an output shaft sub-link having one side connected to the output shaft link member and the other side connected to the main link member; a first output shaft bearing arranged in a connection area between the output shaft sub-link and the output shaft link member; an output shaft power transmission pin integrally connected to the output shaft link member, the output shaft sub-link, and the first output shaft bearing and transmitting power from a side of the output shaft link member to the output shaft sub-link; and a second output shaft bearing arranged in a connection area between the output shaft sub-link and the main link member.

7. The orthogonal-axis power transmission device of claim 6, wherein the output shaft connection portion further comprises a plurality of output shaft stop rings provided to fix a position of the output shaft sub-link or the output shaft power transmission pin.

8. The orthogonal-axis power transmission device of claim 3, wherein the input shaft connection portion and the output shaft connection portion have a same structure.

9. The orthogonal-axis power transmission device of claim 2, wherein the main link member comprises: a disc type link body manufactured in a disc shape and in which first and second link connection portions, to which the input shaft link member and the output shaft link member are respectively connected, are formed on both sides in opposite directions; at least one weight reducer formed in a center area of the disc type link body by passing therethrough, and reducing weight of the disc type link body; and at least one inertia moment balancer provided in side regions of the disc type link body adjacent to the input shaft link member and the output shaft link member to balance moment of inertia during a rotational motion of the disc type link body.

10. The orthogonal-axis power transmission device of claim 2, wherein the main link member is an S-type link member in an S shape, and the input shaft link member and the output shaft link member are provided in a same shape as a Y-type link member in a Y shape.

11. The orthogonal-axis power transmission device of claim 2, wherein the input shaft link member is key-coupled to the input shaft assembly by an input shaft key, and the output shaft link member is key-coupled to the output shaft assembly by an output shaft key.

12. The orthogonal-axis power transmission device of claim 1, wherein the main structure assembly comprises: a structure body in which the space portion is formed, and on one side of which an input shaft assembly connection portion, to which the input shaft assembly is connected, is formed; a structure cover on which an output shaft assembly connection portion, to which the output shaft assembly is connected, is provided, the structure cover being connected to the structure body; and a plurality of structure cover fastening members detachably fastening the structure cover and the structure body.

13. The orthogonal-axis power transmission device of claim 12, wherein the main structure assembly comprises: a structure cover O-ring arranged between the structure cover and the structure body and hermetically sealing between the structure cover and the structure body; a structure cover dowel pin arranged between the structure cover and the structure body and aligning a position of the structure cover with respect to the structure body; a front cover shielding one side front opening of the structure body; a plurality of front cover fastening members detachably fastening the front cover to the structure body; and a front cover O-ring arranged between the front cover and the structure body and hermetically sealing between the front cover and the structure

body.

14. The orthogonal-axis power transmission device of claim 1, wherein the input shaft assembly comprises: an input shaft structure connected to the one sides of the main structure assembly; a plurality of input shaft fixing bolts to fix the input shaft structure to the main structure assembly; an input shaft adapter rotated by the motor, having one side connected to the link assembly for changing a power transmission direction, and inputting the rotational power of the motor to the link assembly for changing a power transmission direction; an input shaft adapter bearing arranged in the input shaft structure and supporting a rotational motion of the input shaft adapter; and an input shaft adapter bearing preload plate forming a preload on the input shaft adapter bearing.

15. The orthogonal-axis power transmission device of claim 14, wherein the input shaft assembly comprises: a motor installation plate on which the motor is mounted; a clamp fastening a motor shaft of the motor to the input shaft adapter; and a clamp tightening bolt tightening the clamp.

16. The orthogonal-axis power transmission device of claim 1, wherein the output shaft assembly comprises: an output shaft structure connected to the other side of the main structure assembly; a plurality of output shaft fixing bolts fixing the output shaft structure to the main structure assembly; an output shaft adapter connected to the other side of the link assembly for changing a power transmission direction and outputting power transmitted from the link assembly for changing a power transmission direction; an output shaft adapter bearing arranged in the output shaft structure and supporting a rotational motion of the output shaft adapter; and an output shaft adapter bearing preload plate forming a preload on the output shaft adapter bearing.
